



माधव प्रौद्योगिकी एवं विज्ञान संस्थान, ग्वालियर (म.प्र.), भारत
MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR (M.P.),
INDIA

Deemed University
(Declared under Distinct Category by Ministry of Education, Government of India)
NAAC ACCREDITED WITH A++ GRADE



Event Report

ALL INDIA SEMINAR

ON

**Prioritizing Safety Culture Across Aviation, Aeronautics, Astronomy &
Space Technology**

Organized By:

Department of Mechanical Engineering

Student's Chapter The Institute of Engineers India

and

Aerospace Club, MITS-DU Gwalior

Event Date: 31st OCTOBER 2025



1. Event Overview

On October 31, 2025, MITS Gwalior hosted a historic dual-event featuring the 1st National Seminar on Safety Culture based on the 1st National Seminar on Safety in Aviation, Aerospace and Space Technology as well as an Aerospace Club Model Purpose Day.

The National Safety Culture Seminar was designed to be an opportunity to discuss the safety culture of the aviation, aerospace, and space technology community, including safety practices, emerging technologies, and the importance of developing and maintaining a strong safety culture within those disciplines. The Model Presentation Day provided an opportunity for students to showcase their innovative, technical, and creative projects ranging from conceptual digital models, to physical prototypes, as well as showcase their ability to apply the skills of solving problems through the use of engineering principles.

The program was diverse in nature and included:

- The presentation and keynote address by esteemed professionals and educators as speakers
- Student-led demonstration of their technical models, innovative designs, and functional concept models
- Presentation of research papers that provide opportunities for students, faculty, and researchers to share ideas and exchanges related to research
- Interactive sessions between students and faculty
- Media coverage that documented key events during the program and could be used to promote the event

Through these combined efforts, a unique opportunity will have been provided to establish an educational and hands-on learning environment that supports the overall development of aerospace education and creates an atmosphere for innovation and safety culture.



2. Objectives

- **Promoting Safety Culture:**

Emphasize the critical importance of safety practices and awareness across diverse domains including aviation, aeronautics, astronomy, and space technology, thereby fostering a culture of responsibility and precaution in both academic and professional settings.

- **Providing a Research Platform:**

Create an inclusive forum for researchers, academicians, and students to present and discuss their work on cutting-edge topics such as safety innovations, artificial intelligence, machine learning applications, and space weather studies.

- **Showcasing Student Innovation:**

Highlight student-built aerospace models that demonstrate creativity, technical proficiency, and practical problem-solving, encouraging experiential learning and hands-on engagement with real-world challenges.

- **Strengthening Collaboration:**

Facilitate meaningful interaction between industry and academia, inspiring cross-disciplinary collaboration and knowledge exchange that can drive innovation and future advancements in aerospace and related fields.

3. Model Presentation

The **Aerospace Club** conducted a dedicated *Model Presentation* session to showcase the hard work, innovation, and technical skills of four student teams. Each team designed, built, and tested an aerospace model following real-world aeronautical principles. Below is the detailed information about each team and their respective models.

Team Name	Model Presented
Team Noctivian	Single-Stage Water Pressure Rocket
Team Aetherion	“Atom” Fixed-Wing RC Aircraft
Team Velocitra	Multi-Rotor Drone System
Team Pyroxis	Solid-Fuel Sugar Rocket

4. Conference Summary

During the session, **students presented research papers on diverse and fascinating topics**, reflecting their innovation, creativity, and curiosity across aerospace, astronomy, and aviation technology.

S. No.	Research Paper Title	Presented By
1	<i>Artificial Intelligence in Aviation Safety: A Review</i>	Osheen Dohare (CSE, 3rd Year)& Ashish Kumar Sahu(AI, 3rd Year)
2	<i>Beyond Known Faults: A Few-Shot Paradigm for Safer Autonomous Space Operations</i>	Ocean Agarwal (CSE, 3rd Year)
3	<i>Object Detection for Safety in Orbital Station Environments Using YOLOv8: Towards ISRO's Bharatiya Antariksh Station Vision</i>	Shreya Goyal (CSE, 2nd Year)& Aneeka Dwivedi (MAC, 2nd Year)
4	<i>Impact of Safety Culture on Reducing Human Error in Aviation Operations</i>	Pihu Mehto (IOT, 2nd Year)& Yash Pal (IOT, 2nd Year)
5	<i>Solar Flare Prediction and Intensity Estimation Using a Transformer Encoder with Multi-Head Cluster Attention</i>	Pulastya Bhagwat (ETE, 3rd Year)& Lavin Panchgavkar (ETE, 3rd Year)
6	<i>The Role of Supermassive Black Holes in Galaxy Formation and Evolution</i>	Shubhi Agrawal (CSE, 3rd Year)

These presentations showcased the depth of student research, their ability to connect theory with practice, and their passion for contributing fresh perspectives to the scientific community.



5. Acknowledgements

We take this opportunity to extend our heartfelt gratitude to all those whose contributions made this seminar and model presentation a resounding success:

- **Special Guests and Speakers**

We extend our heartfelt gratitude to **Er. R. K. Khetan Sir (Chairperson), Prof. K. N. Jha Sir (Co-Chairperson), and Er. Morison Maurya Sir (Programmer Analyst, Boeing Pvt. Ltd.)** for sharing their expertise, delivering inspiring talks, and motivating students to pursue innovation with responsibility and vision. Their presence added immense value to the seminar and presentations.

- **IEI Gwalior Local Chapter**

For their initiative and leadership in organizing the National Seminar, providing a platform that brought together academia, industry, and students to engage in meaningful dialogue on aerospace safety and innovation.

- **Faculty Members**

Dr. C.S Malvi Sir, Dr. Nitin Upadhyay Sir, and Dr. Neeraj Mishra Sir for their constant guidance, encouragement, and valuable insights throughout the planning and execution stages. Their support ensured the academic rigor and smooth conduct of the programme.

- **Student Coordinators and Aerospace Club Volunteers**

For their tireless efforts in managing logistics, coordinating sessions, and assisting participants. Their dedication and teamwork were instrumental in ensuring the seamless flow of the event.

- **Participating Teams**

For their creativity, professionalism, and commitment in preparing and presenting innovative aerospace models. Their contributions exemplified the spirit of experiential learning and technical excellence.

6. Glimpses of Event



7. Posters of Model Presented

WATER ROCKET-TEAM NOCTIVIAN

Physics behind the model

A water rocket operates on Newton's Third Law, where compressed air expels water to create upward thrust. Its fins provide stability, while a balanced air-water ratio ensures maximum height. The streamlined body reduces drag for smoother flight.



Key evidence

Materials Used:

- 2L plastic bottle (body)
- 0.5-inch diameter PVC pipe (launcher nozzle)
- Bicycle valve (for air injection)
- Adhesive & electric tape (for sealing joints)
- Cardboard fins (for stability)
- Bicycle pump (for pressurizing air)

Working Principle:

Compressed air enters through the valve and forces water out of the nozzle, creating thrust in the opposite direction — a direct application of Newton's Third Law.

Opposition

Some argue that water rockets are too simple to demonstrate "real" rocket science.

However, simplicity is their strength — they mirror the same action-reaction principle used by jet engines and space rockets, but in a safe, scalable form. Their low cost and accessibility make them invaluable for education, prototyping, and public science engagement.

— Demonstrates key aerospace engineering principles through a simple, eco-friendly model.



EROSPACE

Team Atherian RC Plane

Central Thesis

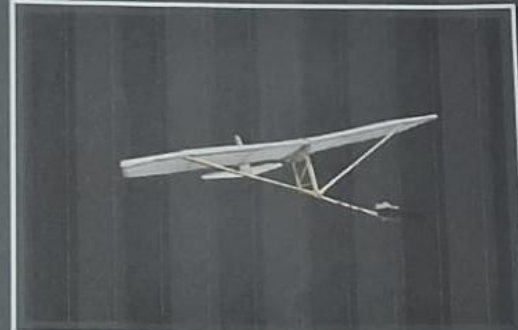
The Team Atherian RC Plane showcases a successful integration of engineering and advanced technologies to achieve exceptional flight performance and innovative design.

Key evidence

- **Aerodynamic Efficiency:** The unique wing and fuselage design minimizes drag, maximizes lift, allowing for extended times stable maneuvers.
- **Structural Integrity:** Constructed from carbon fiber and reinforced foam, ensuring strength while maintaining low overall weight.
- **Advanced Propulsion:** Utilizes a custom-tuned electric motor and reinforced motor system, ensuring power delivery and precise throttle response.
- **Integrated Avionics:** Features a flight control unit with GPS and telemetry feedback, enabling data logging during flight.
- **Modular Design:** Components are easy to assemble, repair, upgrade, enhancing user experience, and longevity.

Call to action

Join the Atherian team! Learn more about our designs, join us, and be part of the next generation RC aviation. Visit atherianrc.com



Paraphrase or use flow quotes to condense and maximize words

"The triumph of engineering." - Atherian Design Team

"Engineered for future flight."

- RC Tech Journal

Opposition

While some argue for traditional balsa wood construction, composites offer superior strength-weight and durability, crucial for performance RC flight.

Sources

RC Flight Engineering Blog, "Atherian RC Plane Review" Fall 2023
Online Community: RCUniverse, WattFly. Discussions: Atherian designs.



EARLY QUADCOPTER DESIGN

A MODEL WORK BY TEAM VELOCITRA

MATERIAL

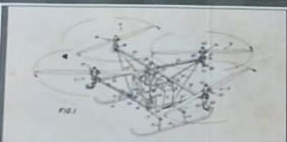
We have used pen body as frame material it gives rigt yet durable and light weight frame which is the most essential parameter in building drones and quadcopters

Specifications

- 1) corless dc motors 3.7v rated rpm 48k
- 2)teflon wires for connection and light weight for weight maintainance
- 3) li- ion 2000mah 3.7-4.2v rated voltage, we used 2 of them which adds up the voltage $\times$ 2 times of both in series combination
- 4)propellors two fins and 2 of them are dockwise and 2 are anti clockwise spin, for mantaining the torque

WORKING PRINCIPLE

Our drone demonstrates basic hovering by balancing upward thrust and weight using four motors—two rotating clock wise (CW) and two counterclockwise (CCW)—to cancel rotational torque. Without Arduino control, the motors were directly connected to a battery for simple lift. Copper rods were added to balance the center of mass, enabling brief stable hovering and proving the principle of lift and torque cancellation in a basic quadcopter setup.



EROSPACE



SUGAR ROCKET

TEAM
PYROXIS

Core Idea:-

The core idea is to produce controlled thrust for low-cost, educational rocketry by using a sugar based solid propellant inside a safely designed motor.

Physics Behind it:-

- Basic Principle: Newton's 3rd Law of Motion
> "For every action, there's an equal and opposite reaction."
- Law of inertia: Newton's 2nd Law of motion
> " $F = ma$ " -> thrust overcomes gravity

Fuel Composition:-

- sugar: the fuel.
- Potassium nitrate (KNO_3): the oxidizer.
- Ratio: usually ~75% KNO_3 to ~25% sugar by weight.

Materials Used:-

- CPVC pipes.
- cardboard
- sugar 25%
- KNO_3 75%



8. List of Students Attended:

SERI NO.	NAME
1	Varnit Kumar
2	Pihu Mehto
3	Charu Saxena
4	Pushkar Agarwal
5	Parth Tomar
6	Bhoomi Jain
7	Kashish sharma
8	Mayank Dwivedi
9	AAHAN SINGH
10	Prateek Sharma
11	Divyanshi Gupta
12	Tanisha kulshrestha
13	Sonam Sharma
14	Mayank Dwivedi
15	Om Mishra
16	Ocean Agarwal
17	Lavin Panchgavkar
18	Diya Singh
19	Shubhra Singh
20	Osheen Dohare
21	Gaurang Sharma
22	Utkarsh singhal
23	Manya Sharma
24	Devanshu Gupta
25	Mehak Kapoor
26	Shreya Goyal
27	Rishabh singh
28	Pulastya Bhagwat
29	Rishabh singh
30	Khushi Agarwal
31	Lakshya Ghanghoriya

- 32 Varennyam Joshi
- 33 Tarun Adhwaryu
- 34 Nitesh Patidar
- 35 Ashish Kumar Sahu
- 36 Yash pal
- 37 Shubhi agrawal
- 38 Aneeka Dwivedi
- 39 Harshita soni
- 40 Aryan Chourey
- 41 Aniya Jain
- 42 Krish Mathur
- 43 Shivam Sharma
- 44 Abhishek Gupta
- 45 Shubash Dangi
- 46 Ananya Dubey
- 47 Aditya Mishra
- 48 Arnav Upadhyay
- 49 Aryan Sharma
- 50 Manjeet rajput
- 51 Naitik goyal
- 52 Prabhanshu kamal ahirwar
- 53 Guru tiwari
- 54 Saksham Sharma
- 55 Prashant sen
- 56 Love mishra
- 57 Vikas Dhakad
- 58 Mahi gupta
- 59 Akmal khan
- 60 Akshat Soni
- 61 Nikhil singh tomar
- 62 Om nema
- 63 Sohail khan
- 64 Udipt kushwaha
- 65 Adi Jaiswal

- 66 Pragya singh
- 67 Mahima Sutrakar
- 68 Rahul yadav
- 69 Mohit Sharma
- 70 Prajwal saki
- 71 Saksham sarang
- 72 Shahid Ansari
- 73 Shivam baghel
- 74 Shivam Uikey
- 75 Shubh Yadav
- 76 Aanjenya Pratap Singh
- 77 Abhinav Cheepa
- 78 Shubham Prajapati
- 79 Anuj Savita
- 80 Dev Srajan Purohit
- 81 Kopal Shah
- 82 Divyanshu Pandey
- 83 Mihika Singh Thakur
- 84 Pragati Chauhan
- 85 Pratham Gupta
- 86 Saurabh Jain
- 87 Soumya Singh Kushwaha
- 88 Swades Kumar
- 89 Prakhar Nema
- 90 Himanshu Pippal
- 91 Anusha anu Prasad